

The ICT Lounge

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Year 7

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IGCSE

A-Level

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Section 4.8: Setting up a Small Network

In this section we will look at how you can **set up** and **configure** a **small network** of your own.

Key Concepts of this section:

- # Understand that you will need both **hardware** and **software** to create a **network**.
- # Know the different **hardware** and **software** needed to create the **network**.
- # Be able to explain the **steps needed** to create a small network.

How to set up and configure a small network

To set up a **small network** (of say 5 computers) that can **share** an **internet connection**, you would need the following:

- [Correct hardware](#)
- [Correct software](#)
- [An internet service provider \(ISP\)](#).

NOTE:

If you were trying to set up a **larger** network, you would need to use more **sophisticated** hardware and software than the types we are going to look at.

We will look at all of these and explain where they fit into the network:

What hardware will I need?

- # The hardware that we will need is listed below:
- **Computers** or **devices** (laptops, desktop PC, tablets etc.)
 - Other **network devices** (printers scanners etc.)
 - A **router/modem** (so computers on the network can send data to each other)
 - **Network interface cards** (wired or wireless)
 - Network **cables** (so any devices with wired NIC's can plug into the router)
 - **Telephone line/Cable line** (to connect the network to a WAN/internet).

What software will I need?

- # The software that we will need is listed below:
- A **firewall** (to help protect against hackers)
 - **Anti-virus** (to help keep the network virus-free)
 - A **web browser** (so computers connected to the network can use the web)
 - An **email** account.

What is an internet service provider (ISP)?

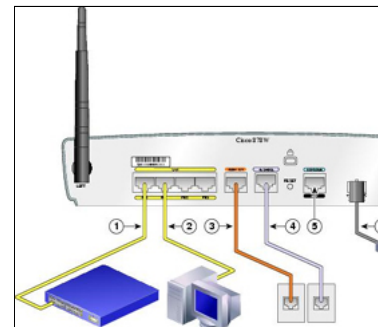
- # An ISP is a company that gives you **access** to the **internet**.
- # Internet service providers will charge you a **monthly fee** in return for internet access.



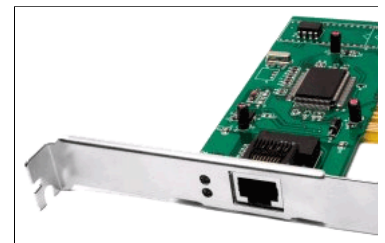
Key Words:

Wi-Fi, Bluetooth.

Examples:



A router will allow you to network all of your devices and share an internet connection.



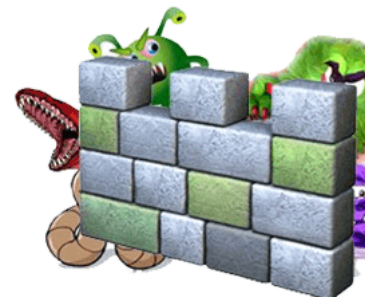
A network interface card will let you connect all devices to the router.



A firewall is software that protects your device from potential threats on the internet.

The **amount** you **pay** will vary depending **how fast** you want your connection to be.

Popular ISP's here in Bermuda include **Logic** and **CableVision**.



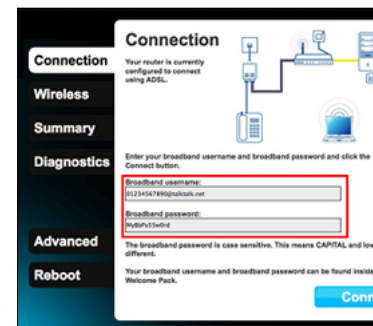
Anti-virus software will shield your networked devices against viruses.

How does all of this fit together?

Once you have all of the hardware/software and an internet service provider, creating the network is very straightforward.

The table below explains how:

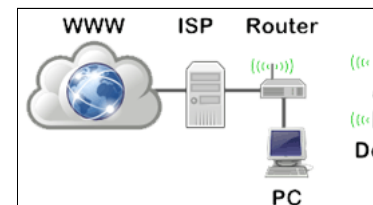
1. Sign up and pay for an account with an **internet service provider**.
(Your ISP will usually supply you with a router/modem)
2. Connect the **router/modem** to one end of the **telephone line** or the **cable line**.
Plug the other end of the line into your **telephone/cable box**.
3. Connect **cabled devices** to the router using your **network cable**.
(This network cable will plug into the network interface card on the device and the ports on the router)
4. Connect **wireless devices** to the router using **Wi-Fi**.
(For each device, you will need to configure their settings and enter the password to access the router. Password is supplied by the ISP)
5. Install and configure a **firewall** and **anti-virus** software on each device that will form the network.
(These will protect your networked devices against hackers and viruses)
6. Configure the **router** with your **internet login details** supplied by the ISP.
7. Use the **web browser** installed on each of your networked devices to access the internet.



You access the internet by entering your ISP details into the router.



Access the internet through your preferred browser.



And there you have it.... a small network!

Activity!

[Setting up a small network - Research Sheet](#)

Click the above task and answer the questions about how to set up a small network.



[Previous - Wi-Fi and Bluetooth in networks](#)

Next -



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